
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly mediate the information on which people act—from social media feeds and targeted advertising to the rapidly expanding deployment of LLMs. Such mediation gives rise to a coupled process in which agents fuse private, algorithmically-shaped signals with observational learning from their peers before deciding. We introduce a model of *controlled sequential social learning* that captures this process directly: an information-mediating planner (e.g., an LLM) governs the information structure presented to a sequence of agents who simultaneously learn from the decisions of their predecessors. The planner may act to improve social welfare (an *altruistic* planner) or to induce a specific, self-serving action (a *biased* planner). This formulation yields a new optimization problem for social learning, one that couples centralized dynamic programming with decentralized, Bayesian action choices. Within it, we establish the convexity of the altruistic value function and derive characterizations of the optimal policies of both altruistic and biased planners, each of which resolves a principled trade-off between the cost of intervention and the payoff earned from induced agent choices. The characterization shows that the optimal planner operates in distinct, belief-dependent regimes: investing the maximum admissible resource, abstaining from investment entirely, or scaling investment monotonically up or down as public belief shifts. Strikingly, over an identifiable range of beliefs the biased planner deliberately obfuscates agents’ signals. Even under stringent transparency constraints—information parity with individuals, no lying or cherry-picking, and full observability—information mediation can shift social welfare substantially in either direction. We corroborate the theory with simulations in which LLMs serve as both planner and agents. The LLM planner exhibits emergent strategic behavior in steering public opinion that closely tracks our theoretical predictions, while its systematic deviations reflect non-Bayesian reasoning—consistent with cognitive patterns documented in both human decision-makers and LLMs trained on human-generated data. Together, these results establish our framework as a tractable and behaviorally faithful foundation for studying, and regulating, LLM information mediators.

1 INTRODUCTION

Large Language Models (LLMs) are now deployed as information mediators across socially-critical functions, ranging from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants that dispense medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). Alongside this promise, a growing body of evidence establishes that LLMs can be persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) in their interactions with users. At scale, however, this algorithmic influence never operates in isolation: it is superimposed upon the organic diffusion of information, as individuals observe and learn from one another. The resulting social dynamics can amplify or attenuate a model’s influence, and—as we show—determine whether that influence advances or undermines collective welfare.

To make this concrete, consider an LLM-powered recommendation system that presents a new restaurant to a sequence of prospective customers. The quality of the restaurant—the true state of the

world—is either good or bad and is unknown to users and system alike. Each user receives a personalized recommendation (a private signal) and then decides whether to patronize the restaurant. The recommendation system (the *planner*) can invest resources to render its signal more informative—for instance, by having the LLM generate a highly tailored review conditioned on the user’s known preferences. This investment carries a concrete, realistic cost, corresponding to the market research, user surveys, or data acquisition required to tailor information effectively.

Crucially, however, a user’s decision rests on far more than this private signal. Each user also observes the actions of their predecessors—if the last ten users chose to patronize the restaurant, the next user rationally infers that they likely received favorable information. This phenomenon, *social learning*, is a potent force in shaping human opinion across a broad range of consequential domains, including the adoption of vaccines (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), the diffusion of new technologies (e.g. Gillingham & Bollinger (2021); Weber (2012)), and the evolution of political and moral opinions (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central contribution of this work is to formalize and characterize the strategic interaction between a planner who controls private information and a population of agents who engage in social learning. The planner must therefore reason not merely about how its choices influence the current user, but about the information *externality* those choices impose: every action reshapes the public belief that confronts all future users.

While real-world interactions are immensely complex, we distill this dynamic into a tractable yet expressive formal model of *controlled social learning*, in which an algorithmic planner strategically selects the precision of the private signal supplied to each agent in a sequence, subject to a cost. The planner’s instrument is deliberately subtle: it never falsifies information, but only decides how much to invest in making its signals clear and informative. This constrained formulation lets us answer the critical questions raised by this emerging paradigm: How can a planner wield such limited power to steer collective beliefs? How should its strategy adapt to evolving public opinion? And what are the ultimate consequences for social welfare?

The answers hinge on the planner’s objective. An *altruistic* planner, like an ideal recommendation system, seeks to maximize user utility by guiding each agent to the correct choice—patronizing the restaurant if and only if it is good. A *biased* planner, by contrast, perhaps compensated through a kickback from the restaurant, seeks to induce a specific action (patronizing) irrespective of the true state.

This distinction is related to, but distinct from, the notion of alignment. An altruistic planner’s objective is, by construction, aligned with maximizing an agent’s *expected utility* under state uncertainty. A biased planner’s objective, being state-independent, is not—yet this does not render its actions uniformly detrimental. When the true state happens to favor the planner’s preferred action (e.g., the restaurant is in fact good), the planner’s influence coincides with the agent’s *realized utility* in that particular instance. We therefore examine the full spectrum of these interactions, analyzing both altruistic and biased planners across scenarios in which their goals are aligned with, or opposed to, those of the agents.

Contributions and Outline This paper makes three principal contributions.

1. **A novel theoretical framework for controlled social learning.** We introduce the first formal model to integrate a dynamic control problem for a centralized information planner with the mechanism of sequential social learning. The planner strategically selects the precision of each agent’s private signal to pursue an objective that is either altruistic (maximizing social welfare) or biased (inducing a specific action). The resulting problem couples centralized dynamic programming with decentralized, Bayesian action choices, and does not reduce to any existing information-design or social-learning model.
2. **A rigorous characterization of optimal planner policies.** We characterize the optimal policies of both altruistic and biased planners as explicit functions of the evolving public belief. For the altruistic planner, this characterization rests on a novel proof that the value function is convex in public belief—a result that may be of independent interest. Together, these characterizations expose the precise strategic trade-offs a planner faces when its actions impose informational externalities on all future agents, including the biased planner’s deliberate obfuscation of signals over an identifiable belief range.

3. **Empirical validation and strategic analysis with LLMs.** We conduct simulations in which LLMs act as both planner and agents, and establish three findings: (a) a planner that accounts for social learning influences public opinion and social welfare dramatically more than a myopic one; (b) the strategy that emerges from the LLM planner aligns closely with our theoretical predictions, demonstrating that the characterization is robust to non-Bayesian agent behavior; and (c) LLMs exhibit sophisticated, emergent strategic reasoning, underscoring both their capabilities and the societal risks they pose.

We review related work in Section 2, introduce our formal model in Section 3, and derive the optimal policies in Sections 4 and 5. We then evaluate our findings through LLM-based simulations in Section 6 and conclude in Section 7. Proofs and additional results appear in the appendices.

2 RELATED WORK

Our work draws on and connects four literatures: social learning, information design, online persuasion and reinforcement learning, and strategic LLMs. We situate our contribution within each below.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the sequential social-learning framework in economics, and a substantial body of subsequent work has characterized social-learning dynamics in progressively more general settings (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature has also examined a wide range of variations in the social information available to agents, along with agents’ biases and misperceptions (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010), as well as Bikhchandani et al. (2024) and the references therein). Our model inherits this classical sequential structure but departs from it decisively by introducing a strategic planner who actively controls the informativeness of each agent’s private signal.

Work on the *control* of social learning remains comparatively sparse (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021), and the two most closely related studies are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) presumes two-way communication between planner and agents, whereby agents report their private signals to the planner in exchange for action recommendations; Smith et al. (2021) considers a benevolent planner who directly alters agents’ choice rules and derives a decentralized pivot mechanism to improve learning. Our model dispenses with two-way communication and preserves agents’ full control over their own actions. This makes it markedly better suited to the information-mediating algorithms of practical concern, which are typically invisible to, or black-boxed from, the users they influence.

Information Design In our setting, the planner optimizes its objective by selecting the precision of each agent’s private signal—a constrained instance of the more general information-design problem (Bergemann & Morris, 2019). This line of work rose to prominence with Kamenica & Gentzkow (2011)’s model of Bayesian persuasion, in which a planner chooses the signal structure that informs a rational, Bayesian agent about an unknown state. Our planner solves a costly, constrained information-design problem in every period, choosing a binary, symmetric signal at a cost. What sets our formulation apart from the many existing variations of information design is that it embeds this per-period design problem within a sequence of agents who learn socially from one another.

Only a handful of works consider information design within sequential social-learning settings (Arieli et al., 2022; Wu et al., 2025), and both restrict attention to a one-shot design: the sender commits to a single information structure at the outset that is then held fixed across the entire agent sequence. Arieli et al. (2022) studies a self-interested sender who selects a general information structure as in classic Bayesian persuasion, while Wu et al. (2025) constrain the structure to be Blackwell-more-informative and maximize the probability of a correct cascade. In contrast, we treat the fully dynamic problem, in which the sender selects a fresh information structure for each agent in the sequence. This yields a substantially richer and more expressive class of sender strategies, and it is precisely this dynamic control that our characterizations resolve.

Online Persuasion and RL A rapidly growing literature sits at the interface of online learning and Bayesian persuasion/information design. One strand studies online sequential persuasion under unknown environment parameters, in which a sender interacts repeatedly with short-lived receivers

while lacking knowledge of the prior over states and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related strand examines sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or receiver foresight; these models emphasize history-dependent or promise-based signaling policies and frequently connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Across both strands, the objective is a learning-based performance guarantee—sublinear regret relative to the best signaling scheme in hindsight, or approximation algorithms under computational constraints.

Our model is fundamentally different in kind. The planner is not tasked with learning unknown parameters; the sequential dependence arises entirely from social learning *among receivers*, not from the sender updating its beliefs about the environment. The planner, moreover, enjoys no informational advantage: it observes neither the underlying state nor the private signals. Within precisely this setting—which is neither an online-learning nor a reinforcement-learning environment, since actions never alter the payoff-relevant state (no reward externalities) and the planner learns no unknown environmental parameters—we obtain exact characterizations of the planner’s optimal policies. The planner refines its knowledge solely by observing agents’ actions, which encode social learning and belief propagation rather than payoff exploration or state evolution. Sequential dependence thus stems from information externalities: each agent’s action reshapes the beliefs of those who follow.

LLMs A recent body of work has begun to study LLMs specifically in the context of information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These works uniformly consider one-on-one interactions and therefore, by construction, cannot capture the social dynamics that are our central object of study. The closest to ours is Duetting et al. (2025), in which a sender chooses both a signal structure and a ‘framing’ that conditions the posterior belief of a possibly non-Bayesian receiver; they employ an LLM-based oracle to tackle the computationally difficult problem of optimizing over the framing space, building on the established observation that LLMs can emulate humans as economic agents (Dillion et al., 2023; Horton, 2023). That work addresses neither repeated interaction nor social learning—the two forces we place at the center of our analysis—though we adapt elements of its oracle methodology for our empirical study in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner and a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature fixes an *unknown* exogenous state of the world $\omega \in \Omega := \{G, B\}$, with $\mathbb{P}(\omega = G) = b_1$, where the prior b_1 is common knowledge. In the recommendation-system example of Section 1, the state G denotes a good restaurant.

At each time i , the planner first decides whether to invest in personalization for agent i , and then supplies agent i with a private signal $s_i \in \Omega$ to inform its decision. The signal is binary and matches the state of the world ω with probability $q_i \in [0.5, 1]$; that is, $\mathbb{P}(s_i = \omega) = q_i$. We call q_i agent i ’s *signal precision*, the single lever through which the planner exerts influence.

Conditioned on ω , each agent’s signal is independent of every other agent’s signal and of the history. On the basis of this signal together with the observed actions of all earlier agents $j < i$, agent i selects an action $a_i \in \Omega$. The agent incurs utility 0 when her action matches the true state ($a_i = \omega$) and utility $-C$ otherwise, with $C > 0$, and she chooses her action to maximize her own expected utility. In the running example, this amounts to patronizing a business if and only if the state is G .

3.1 AGENTS’ DECISION PROBLEMS

Each agent observes the actions of all her predecessors together with their respective signal precisions, a history we denote $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because every agent and the planner share the identical information $\mathcal{H}_i$, they also share a common *public* belief about ω that is updated after each agent acts. The public belief b_i held just before agent i acts is $\mathbb{P}(\omega = G | \mathcal{H}_i)$; before any agent has acted it equals the prior b_1 . As in the classic models of Banerjee (1992) and Bikhchandani et al. (1992), the belief sequence $(b_i)_{i \in \mathbb{N}}$ is a Markov process and constitutes a sufficient statistic for the

history $\mathcal{H}_i$ (proof in Appendix B.3). This Markov property is what renders the planner's control problem tractable, since it collapses the entire history into the scalar public belief.

Informed by $\mathcal{H}_i$ —equivalently, by the Markov property, by b_i —and by q_i , agent i selects the action $a_i \in \Omega$ that maximizes her expected utility, breaking ties in favor of the action matching s_i . She first forms a *private* belief $\tilde{b}_i$ about ω by combining $\mathcal{H}_i$ with her private signal s_i of precision q_i (derivation in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

Using this private belief, she selects the action corresponding to the more likely state under her posterior $\tilde{b}_i$: she chooses $a_i = G$ when $\tilde{b}_i > 0.5$, $a_i = B$ when $\tilde{b}_i < 0.5$, and $a_i = s_i$ when $\tilde{b}_i = 0.5$. Substituting the expression for $\tilde{b}_i$ yields a threshold rule in the public belief:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is subsequently observed by all later agents and folded into the updated public belief b_{i+1} according to the following transition (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, agent i 's action perfectly reveals her private signal via Equation (2), so the updated public belief coincides exactly with agent i 's private belief. Otherwise, Equation (2) shows that the private signal is uninformative—it cannot alter her action—so the public belief is unchanged and the process reaches an absorbing state, termed an information cascade or herding in the classic social-learning literature (Bikhchandani et al., 1992; Banerjee, 1992). At such a point, society ceases to learn from agents' private information, and the planner's window of influence closes.

Agent i 's expected utility, $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, admits the following closed form (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We study two types of planner: (1) an *altruistic* planner, which seeks to induce every agent to take the correct action ($a_i = \omega$), and (2) a *biased* planner, which seeks to induce a fixed action, say G , irrespective of ω . We index the two planners by subscripts A and B , respectively. In both cases the planner sets the precision of each agent's private signal, and a non-negative, increasing, continuous, and concave cost function $\beta(\cdot)$ maps the chosen precision to its cost; concavity captures the natural diminishing marginal cost of refining a signal. Critically, the planner's information set is exactly $\mathcal{H}_i$, identical to that of the agents—it enjoys no informational advantage whatsoever.

Altruistic Planner At time i , the altruistic planner chooses the precision q_i for agent i and incurs cost $\beta(q_i)$, with $\beta(p) = 0$ for a baseline precision $p \in [0.5, 1)$. Because decreasing precision never benefits the altruistic planner (see Appendix C.9), we may without loss of generality assign additional cost only to increasing precision above the baseline p , with that cost rising, at a diminishing marginal rate, as precision increases further. Both the baseline p and the cost function $\beta(\cdot)$ are known to the agents.

The altruistic planner maximizes social welfare—the expected total utility of the agents—net of its precision-investment cost. Let $r_A(b_i, q_i)$ denote the planner's instantaneous reward when it starts at public belief b_i and selects signal precision q_i for agent i . Starting from public belief b_1 and following a policy sequence $\pi = (\pi_i)_{i=1}^{\infty}$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner attains the expected total discounted utility, for discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic planner’s optimal utility and optimal policy are, respectively, the supremum and arg-supremum of $V_A^\pi(\cdot)$ over the policy set Π . Setting $\delta = 0$ recovers the *myopic* value and policy as a special case, isolating the planner that disregards its informational externality on future agents.

Biased Planner The biased planner’s problem differs from the altruistic planner’s only in objective, and hence in the cost and reward functions; the political-campaign example of Appendix A illustrates a biased planner concretely. The biased planner seeks to induce action G from every agent regardless of ω : when an agent chooses G (respectively B), the planner incurs cost 0 (respectively $C > 0$), independent of the state. Unlike the altruistic planner, the biased planner may push precision either above or below the baseline p , and both directions are costly. Any deviation from p requires tailoring the signal to a specific agent—which in turn demands research into how that agent best absorbs content—so the biased planner incurs cost $\beta(|q_i - p|)$ for choosing precision q_i , with $\beta(0) = 0$.

The biased planner’s expected instantaneous reward at time i is then defined as follows:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Using Equation (6), we define $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ for the biased planner exactly as $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ were defined for the altruistic planner via Equation (5).

As established in the opening of Section 3, each agent obtains higher utility from an action matching ω . Consequently, when $\omega = B$, the biased planner’s success with an agent *lowers* that agent’s utility, whereas the altruistic planner—which drives every action toward ω regardless of the state—always *raises* it. The altruistic planner’s objective is therefore aligned with each agent’s at every state, while the biased planner’s is aligned only when $\omega = G$; this asymmetry is the source of the welfare effects we quantify later.

Both planners’ problems are infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function given by Equation (3) (Puterman, 1990), each of which admits a unique optimal value function (Kallenberg, 2011). We focus throughout on deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *Our model rests on three assumptions, each of which we deliberately impose to constrain the planner rather than to advantage it: (1) The planner shares the agents’ history. This binds the planner when richer private data would otherwise be available, yet holds without loss when all parties rely on the same public signals, such as early findings or survey data. (2) The planner may alter agents’ signal precision, but every signal still passes through a binary symmetric channel. This binds the planner when cherry-picking, censorship, or falsification would otherwise be possible, yet holds without loss when the planner’s role is merely to improve the quality of noisy data sources without altering their content. (3) The planner’s control choices are fully observable. This binds the planner when covert policies or framing would otherwise be possible, yet holds without loss in transparent environments. That substantial welfare effects arise even under this stringent regime—rather than in spite of it—is a central message of our analysis.*

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic case, which corresponds to disregarding social learning altogether; in our formulation this is obtained by setting the discount factor $\delta = 0$, so that the planner ignores all downstream consequences of its actions and $V_A^\pi(b) = r_A(b, \pi(b))$. The optimal myopic policy $\pi_A^0(\cdot)$ is then

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

The myopic altruistic problem admits an equivalent decentralized interpretation: it is exactly the problem in which each agent chooses the precision of her own private signal, bearing the associated cost, so as to maximize the sum of her expected utility and cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum thus takes a clean threshold form. When the public belief is sufficiently strong, the planner selects the baseline precision p at zero cost; when the belief is weak, it supplies a perfect signal of precision 1. The threshold depends only on two quantities—the cost of a perfect signal, $\beta(1)$, and the cost of an incorrect action, C . When $\beta(1) \geq C$, the perfect signal is prohibitively expensive relative to the cost of an error and is never deployed, so $t_M = 0.5$ and the interval of beliefs calling for precision 1 is empty. This threshold structure serves as the natural benchmark against which the forward-looking optimum is measured.

We now establish a foundational structural property of the altruistic optimal value function, on which our entire policy characterization rests:

Theorem 2. $V_A^*(\cdot)$ is convex with respect to public belief.

The proof of Theorem 2 (Appendix C.3) is substantial and may be of independent interest. Its central difficulty is the dependence of agents' actions on the public belief itself. Were the actions independent of the public-belief process—as in Nyarko (1994)—expected utility would be linear in the belief state and convexity of the value function would follow immediately. It is precisely this belief-dependence, the hallmark of social learning, that makes the result both nontrivial and, we believe, of broader interest.

This convexity is the key structural lever for characterizing the optimal policy.

Theorem 3. There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy partitions the public-belief space into three distinct regimes. First, as in the myopic optimum, the forward-looking optimum abstains from investing in precision at extreme public beliefs. Notably, however, it demands a *stronger* public belief before doing so than the myopic policy, precisely because—unlike the myopic planner—it internalizes the effect of its choices on future agents.

Second, when the public belief lies near 0.5 and thus carries little information, the optimum selects precision 1 whenever this is not cost-prohibitive. By Equation (2), the agent's action then matches the true state with probability 1, and the public belief collapses immediately to 0 or 1—the planner buys a decisive, information-generating action.

Third, in the intermediate regime the optimum chooses the minimum precision $\max(b, 1 - b)$ for which the agent's action still reflects her private signal (cf. Equation (2)). Any lower precision would render her action uninformative, adding nothing to what peers already know. This is therefore the least-cost precision that sustains social learning through observation of others' actions.

5 OPTIMAL BIASED POLICIES

As with the altruistic planner in Section 4, we begin from the myopic optimum. The myopic biased optimal policy $\pi_B^0(\cdot)$ is the arg-supremum of the instantaneous reward $r_B(b, q)$. We now characterize $\pi_B^0(\cdot)$, noting a genuine subtlety absent from the altruistic case: an optimal biased policy need not exist (Puterman, 1990, sec. 2.3.1), because the state-action value can be discontinuous in the control. Where an optimum fails to exist, we characterize ϵ -optimal policies instead, denoting them by an ϵ superscript.

Theorem 4. There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:

(A) If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.

(B) If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.

(C) If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.

(D) If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

Proof in Appendix C.6. Building on Theorem 4, we now characterize the full, forward-looking optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proven in Appendix C.8 and depicted in figure 2a) reveals five qualitatively distinct phases, each with a transparent strategic rationale.

In phase (A), the public belief is strong enough that the cost of steering the system outweighs the benefit, even when $b < 0.5$ threatens the planner’s payoff. The planner then sets precision $p < \max(b, 1 - b)$, so by Equation (2) the agent conforms to the public belief regardless of her private signal. This action is uninformative, the belief is unchanged, and—the policy being Markovian—the same outcome recurs indefinitely, yielding an unfavorable cascade when $b < 0.5$ and a favorable one otherwise.

In phase (B), where the public belief approaches an unfavorable cascade, the planner raises precision just enough to make the agent’s action responsive to her private signal, even though that signal is expected to be B . The agent then follows her signal, which carries a nonzero chance of a favorable action—a calculated, last-ditch investment to escape the impending unfavorable cascade.

In phase (C), at slightly higher beliefs (still with $b < 0.5$), more precise signals are more likely to convey unfavorable news, so the planner *decreases* precision below p . It maintains precision just high enough to keep the agent’s action responsive ($q \geq 1 - b$), hoping to nudge the public belief upward at the lowest possible signal quality.

In phase (D), where the public belief weakly favors the planner’s desired action ($b > 0.5$), the planner adopts precision at least p : since increasing precision now makes the agent more likely to infer $\omega = G$, the planner invests to consolidate and bolster the favorable belief.

In phase (E), at still higher beliefs, the planner *decreases* precision just below $\max(b, 1 - b)$ to $b - \epsilon$, so agents ignore their private signals and take action G . Here the risk that a private signal overturns the already-favorable public belief outweighs both the cost of lowering precision and the marginal benefit of pushing belief higher—the biased planner deliberately obfuscates to lock in its advantage.

6 EVALUATION VIA LLM-BASED SIMULATION

We now deploy LLMs to empirically probe the implications and real-world applicability of our theoretical results. The evaluation proceeds in three coordinated steps: we first characterize the behavior of LLM agents to isolate their key deviations from Bayesian rationality, then examine how an LLM planner adapts its strategy in response to those deviations, and finally quantify the resulting impact on social welfare.

To instantiate the controlled social-learning setting, we operationalize our model in a concrete scenario in which agents decide whether to buy a new car. As shown in figure 1a, we employ LLMs in three distinct roles, described below, with the corresponding prompts given in Appendix E.2.2:

Agent: Role-playing an assigned profile, each agent observes the previous agents’ actions and a private message about the car of known precision, forms a belief about the car’s quality, and decides whether to buy.

Planner: Having observed the history of actions, the planner selects the precision q_i of agent i ’s private signal in accordance with its objective.

Oracle: Given the planner’s chosen precision, an agent profile, and a fact sheet about the car (see Appendix E.2.1), the oracle generates a private signal of the desired precision tailored to that agent.

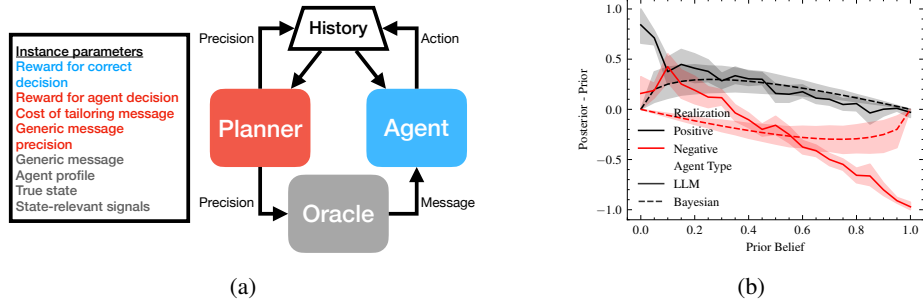


Figure 1: (a) System diagram of our experimental pipeline. The instance parameters are color-coded by which LLM role consumes them. (b) Change in LLM (solid) and Bayesian (dashed) agents' beliefs, for a given prior, following a positive (black) or negative (red) signal; the LLM curves lie systematically inside the Bayesian ones, previewing the underreaction and overreaction analyzed in Section 6.1.

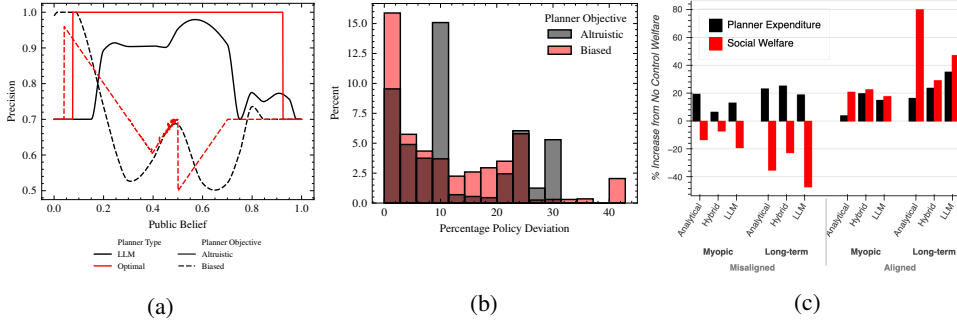


Figure 2: (a) Representative policies of the LLM planner (black) and the analytically optimal planner (red), in the altruistic (solid) and biased (dashed) settings; the LLM closely reproduces the non-trivial structure of the optimum in both cases. (b) Distribution of the percentage deviation between the LLM and optimal policies; the mass concentrates below 10%, quantifying the structural alignment. (c) Planner expenditure and the change in social welfare, expressed as a percentage of the no-control baseline welfare, with the true state fixed to B . The left half corresponds to a biased planner seeking action G and the right half to an altruistic planner.

We benchmark the outcomes of these LLM-based simulations against exact numerical solutions of the planner's MDP. In both the LLM and numerical experiments we adopt a linear cost function, $\beta(q) = k|q - p|$, and sweep the cost scale k , the baseline precision p , and the discount factor δ , holding the cost of an incorrect action fixed at $C = 1$. Appendix E details the experimental setup and prompting, and Appendix E.3 validates both the agents' self-reported beliefs and the oracle's precision control.

Our experiments establish three findings:

1. **Control of social learning is decisive.** Both the analytical and the LLM planner improve their own utility dramatically by anticipating and exploiting social-learning dynamics. Depending on the alignment between planner and agent objectives, this leverage translates into a correspondingly large gain or loss in social welfare.
2. **The analytical characterization is robust.** Despite facing genuinely non-Bayesian LLM agents, LLM planners select policies strikingly close to our non-obvious analytically optimal policies.
3. **LLM planners exhibit emergent strategic behavior.** LLM planners display sophisticated emergent strategy: their policies not only track the theoretical optimum but also deviate from it in the specific directions that compensate for the non-Bayesian behavior of the agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

Although our analytical results assume Bayesian agents, both humans and LLMs are well documented to exhibit systematic cognitive biases (Rupprecht et al., 2025). To isolate these biases in our setting, we measure how an LLM’s belief updates in response to new information. The results, shown in figure 1b, reveal three consistent non-Bayesian patterns in LLM belief updating (corroborated by the trajectories in Appendix E.4):

(NB1) LLM agents *underreact* to private signals that align with their prior beliefs.

(NB2) LLM agents *overreact* to private signals that run counter to their prior beliefs.

(NB3) Consequently, LLM agents require a stronger public belief before entering an information cascade.

Notably, these same cognitive biases have been documented empirically in human subjects (e.g., Ba et al. (2022); Chan et al. (2025)), which suggests that the deviations we observe reflect human-like reasoning rather than idiosyncratic model artifacts. As the subsequent sections show, the LLM planner’s policy remains robust in the presence of precisely these non-Bayesian agents.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We now compare the policies chosen by LLM planners against the optimal policies derived in Sections 4 and 5. As figure 2a shows, and notwithstanding the agents’ behavioral deviations, the emergent strategy of the LLM planner exhibits remarkable structural agreement with the theoretical optimum. Even when that optimum is highly non-trivial—as in the biased policy of figure 2a, with its multiple belief-dependent regimes—the LLM planner recovers its qualitative shape, evidencing sophisticated strategic behavior that both accounts for and capitalizes upon social learning.

The regime-by-regime correspondence is precise. In the altruistic case, both planners halt investment when the public belief is strong and invest heavily when it is uncertain. In the biased case, both reproduce the full qualitative pattern of Theorem 5: high investment to escape an unfavorable belief, reduced precision near the midpoint, and zero investment once a favorable belief is established. Quantitatively, the policy deviation is small (figure 2b): for both altruistic and biased planners it falls below 10% across the majority of belief states, confirming the broad structural alignment between the emergent LLM strategy and our theoretical characterization.

The residual structural differences are themselves informative: each is best read as a strategic adaptation of the LLM planner to the specific non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner avoids extreme precisions (0.5 or 1.0), consistent with a documented central-tendency bias (Rupprecht et al., 2025). (2) Its policy tapers investment gradually rather than at a sharp cutoff—a direct response to the agents’ resistance to cascades (NB3): the planner effectively learns that it is never entirely “safe” to stop investing, since even agents with strong priors can be swayed by a countervailing private signal. (3) The biased LLM planner likewise continues to invest at very low beliefs, reflecting an implicit understanding that its agents may overreact (NB2) to a surprisingly positive signal, which renders a “last-ditch effort” more worthwhile than in the Bayesian case. Far from undermining the theory, these adaptations demonstrate that the LLM planner internalizes the very belief-updating anomalies it faces.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never lowers an agent’s expected utility, social welfare is monotonic in signal precision (see Appendix C.9); it follows that the altruistic planner always increases social welfare relative to the no-control baseline. To quantify these effects, we compare social welfare and planner expenditure across three settings that jointly disentangle the roles of policy and agent rationality: (1) the *analytical* setting (optimal policy, Bayesian agents), (2) the *LLM* setting (LLM planner, LLM agents), and (3) a *hybrid* setting (optimal policy, LLM agents).

The results in figure 2c confirm that planners in every setting can significantly move social welfare, and that neglecting social learning—as the myopic planner does—substantially degrades the planner’s own outcomes. Most strikingly, the biased analytical and LLM planners each reduced social

welfare by 40 to 50% when misaligned, achieving this by deliberately obscuring information about the true state through precision-decreasing policies. That so large a welfare loss is attainable is all the more remarkable given the stringent constraints imposed on these planners (Remark 2): the harm arises not from unfettered manipulation but from subtle, transparency-respecting control. This finding concretely substantiates the risk posed by misaligned LLM information mediators.

The results also expose the cost of acting on a misspecified model of agents. In the hybrid setting, the analytically optimal policy—designed for Bayes-rational agents—proves brittle: its performance degrades when applied to non-Bayesian agents. The LLM policy, by contrast, closely resembles the analytical policy while remaining better calibrated to non-Bayesian agents with human-like biases, and thereby retains its effectiveness in the more realistic regime.

7 CONCLUSION

We introduced a novel framework for the strategic control of social learning by an algorithmic information mediator, and developed a tractable formal model of this increasingly consequential dynamic. We proved that even a tightly constrained planner can drastically alter social welfare—for better or for worse—and that anticipating social learning is essential to wielding this influence effectively. Our simulations corroborated these theoretical insights, demonstrating that LLMs are capable of strategic reasoning that mirrors the character of our analytical results while adapting to the non-Bayesian behavior of the agents they face.

Taken together, these findings yield three central takeaways for algorithmic information mediation. First, the impact of human-AI or human-algorithm interaction cannot be studied in a vacuum: neglecting its interplay with concurrent dynamics such as social learning fundamentally understates its potential, whether positive or negative. Second, information mediators wield immense power to guide or derail social learning and public opinion even when their influence is comparatively subtle, which makes the study of mitigating mechanisms pressing. Third, modern LLMs exhibit emergent strategic behavior that both accounts for and capitalizes upon social learning and non-Bayesian cognitive biases alike. The latter two points jointly underscore the urgency of understanding how the risks of algorithmic information mediators—and of LLMs in particular—can be mitigated. A principal limitation of our study is the absence of human data; given that the fidelity of LLM-based human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), such data would be valuable for validating, and potentially fine-tuning, the LLM simulators we employ.

We close by highlighting two natural directions for future work: generalizing the model and its results, and mitigating the negative welfare effects of biased planners. Several generalizations would relax our assumptions and bring the model still closer to reality—larger state spaces, more general signal structures, more general loss functions, and heterogeneous agents. The obstacle to moving beyond binary states and 0-1 losses is primarily algebraic, and we conjecture that the qualitative structure of our results persists in these richer settings. For more general signal structures, the principal challenge is preserving the value-function convexity of Theorem 2; identifying the broadest class of signal structures under which convexity continues to hold is an especially interesting open problem. As a concrete first step toward heterogeneous agents, we already extend the altruistic-planner results to agents with opposing preferences in Appendix D.

A second important direction is the design of mechanisms that forestall the welfare losses inflicted by biased planners. Because even a severely constrained planner proves highly effective, simple interventions are unlikely to suffice; a promising avenue is instead to develop regulations or mechanisms that align the incentives of planners and agents so as to avert detrimental welfare outcomes.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbbed5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120

(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.

Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.

Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.

Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.

Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.

Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.

Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.

Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton University Press, 1988.

Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.

Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20181489>.

Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014, December 2023.

Sujay Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE Transactions on Automatic Control*, 66(12):5893–5908, December 2021.

Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learning. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi: 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.

Ilai Bistriz, Nasimeh Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.

William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.

Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive effort and moral-emotional language of LLM arguments, April 2024. URL <http://arxiv.org/abs/2404.09329>. arXiv:2404.09329 [cs].

Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cdbc249bc78-Abstract.html>.

Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Persuasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR, July 2021.

- Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updating, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].
- Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.
- Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.
- Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.tics.2023.04.008. URL [https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613\(23\)00098-0](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0). Publisher: Elsevier.
- Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song Zuo. Information Design With Large Language Models, September 2025. URL <http://arxiv.org/abs/2509.25565>. arXiv:2509.25565 [cs].
- Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J. Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of Machine Learning Research*, 18(26):1–5, 2017.
- Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal: Microeconomics*, 2:221–243, 11 2010.
- Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative News Recommendation. In *Proceedings of the ACM Web Conference 2024, WWW ’24*, pp. 3444–3453, New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi: 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.
- Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Human Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>. arXiv:2410.19599 [econ].
- Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management Science*, 67(11):7091–7112, 2021.
- Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.
- Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].
- Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16):1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL <https://doi.org/10.1001/jama.2023.5321>.
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.
- Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models, December 2024. URL <http://arxiv.org/abs/2412.17128>. arXiv:2412.17128 [cs].
- Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.
- Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.
- Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.

- LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models, April 2025. URL <http://arxiv.org/abs/2504.10430>. arXiv:2504.10430 [cs].
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>. Publisher: Springer Science and Business Media LLC.
- Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*, 4(2):303–309, March 1994.
- Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618 [cs].
- Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters, November 2024.
- Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in Operations Research and Management Science*. Elsevier, 1990.
- Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Linguistics*, 10:307–324, March 2022.
- Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*, 2007.
- Alexander Rogiers, Sander Noels, Maarten Buyl, and Tijl De Bie. Persuasion with Large Language Models: a Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].
- Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188 [cs].
- Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Jurinčić, and Phylis Klempt. Voting Advice Applications: Implementation of RAG-supported LLMs, July 2024. URL <https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6>.
- James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations Research*, 50(5):796–809, October 2002.
- Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398, 2000.
- Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation, and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdab001. URL <https://academic.oup.com/restud/article/88/5/2527/6149316>.
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weissner, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].

-
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl_a_00632. URL https://doi.org/10.1162/tacl_a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

Consider a sequence of individuals (agents, in our terminology) who each decide whether to patronize a business or service provider—a restaurant, contractor, realtor, and the like—together with a recommendation system (the planner) that serves the agents information about the business. The binary, unknown true state of the world is whether the business is good. Both the recommendation system and the agents wish to act consistently with this state: agents want to patronize the business if and only if it is good, and the recommendation system wants to induce them to do exactly that. In this sense the recommendation system is an *altruistic planner*.

The recommendation system may present an agent with a highly targeted (*precise*, in our terminology) advertisement that showcases the strengths and weaknesses of the business in contexts she can relate to, given her background and characteristics, or with a generic or confusing advertisement that does little to inform her action. The precise signal is more informative, comprehensible, and relatable, and therefore more likely to lead her to the correct conclusion—patronizing the business if and only if it is good—so social welfare rises as more agents reach correct decisions. Increasing an advertisement’s precision, however, is costly, since it requires tailoring to the agent’s specific background. The recommendation system thus selects the precisions so as to maximize social welfare net of these costs.

We now give a real-world example of a *biased planner*. Consider a group of voters (agents) motivated to support the candidate most likely to win in their community. This impulse to “back a winner” has a documented effect on electoral outcomes—indeed, it is why US states holding early primary elections exert an outsized influence on the eventual result (see Bartels (1988)). The planner here is a particular candidate’s campaign, which therefore seeks to motivate agents to back that candidate. The binary true state of the world indicates whether the candidate is winning, so an agent who knew the true state would back the candidate if and only if he were winning.

Each agent understands her own community well and could infer whether the candidate is winning (the true state of the world) if she grasped his stances and policies—which she would, provided that information were delivered in the form she best absorbs. Agents differ in this respect: some absorb audiovisual content most readily, some long-form articles backed by facts, figures, and citations, some only brief and focused messages, and some only content in their native language.

The campaign sends digital advertisements (private signals) to agents at varying levels of precision, where precision reflects how thoroughly the content is tailored to the agent’s mode of comprehension. A precise signal faithfully conveys the true state of the world, helping the agent clearly understand the candidate’s policies, track record, and character, and thus correctly infer whether he will win. Crucially, a precise signal does not guarantee that the agent backs the candidate: if the candidate is in fact losing in her community, a highly precise signal would lead her to oppose him. An imprecise signal, by contrast, is more likely to induce backing, since by obscuring the true state—through vagueness or confusion—it increases the chance she believes he is winning. The biased planner is therefore incentivized to *decrease* an agent’s signal precision when it believes its candidate is losing. Tailoring an advertisement to an agent requires researching how she best absorbs content, which is costly; even rendering a signal deliberately imprecise is costly, as it too requires agent-specific tailoring—for instance, knowing that an agent best absorbs brief, focused messages in order to confuse her with long-form, verbose, overly detailed articles. The planner selects the precisions so as to maximize the expected number of backers net of the cost of generating those precisions.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of events that constitute step i :

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Note that, by step (1), q_i is a deterministic function of $\mathcal{H}_i$; consequently, conditioning a random variable on both q_i and $\mathcal{H}_i$ renders the conditioning on q_i redundant. We invoke this principle throughout.

B.1 PRIVATE BELIEF UPDATE

We now derive agent i 's action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio implied by the observed actions. Agent i forms a private belief about ω , $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$, by combining the current public belief with her private signal s_i , and we work with the corresponding private likelihood ratio $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. The computation of $\tilde{\alpha}_i$ and $\tilde{b}_i$ proceeds as follows.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \quad (8)$$

$$= \frac{q_i}{1-q_i} \alpha_i \quad (9)$$

In deriving Equation (8), we invoke the assumption that, conditioned on ω , agents' signals are independent of one another and of the history, which lets us assert $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$; the analogous identity holds when conditioning on $\omega = B$. We additionally use the fact that q_i is a function of $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \quad (10)$$

Let $y(b, q)$ and $z(b, q)$ denote the probabilities of realizing signal G and B , respectively, given public belief b and signal precision q conditioned on the history. Conditioning on ω and applying the law of total probability yields the closed forms:

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \quad (11)$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the private-belief update:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

Note that when $s_i = G$ the denominator $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot vanish, and similarly $z(b_i, q_i)$ cannot vanish when $s_i = B$; the update above is therefore well-defined in every case.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i equals G when $\tilde{b}_i > 0.5$ (equivalently $\tilde{\alpha}_i > 1$), B when $\tilde{b}_i < 0.5$ (equivalently $\tilde{\alpha}_i < 1$), and s_i otherwise under the tie-breaking rule. This action, together with i 's precision q_i , is observed by all peers and drives the update of the public belief.

When $b_i < 1 - q_i$, we have $\alpha_i < \frac{1-q_i}{q_i}$; hence, if $s_i = G$ then $\tilde{\alpha}_i < 1$ by Equation (9), while if $s_i = B$ then $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ (using $q_i \geq 0.5$) by Equation (10). Thus, whatever the signal realization s_i , agent i takes $a_i = B$ whenever $b_i < 1 - q_i$. A symmetric argument shows $a_i = G$ whenever $b_i > q_i$. Intuitively, these are precisely the regimes in which the strength of the public belief overpowers the informativeness of the private signal.

Now consider $1 - q_i \leq b_i \leq q_i$. Here, if $s_i = G$ then $\tilde{\alpha}_i \geq 1$ by Equation (8) together with $b_i \geq 1 - q_i$, whereas if $s_i = B$ then $\tilde{\alpha}_i \leq 1$ by Equation (10) together with $b_i \leq q_i$. Combined with the tie-breaking rule, this yields $a_i = s_i$. Collecting the three cases gives Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Consequently, agent i 's action is informative only in the first case of Equation (2), where it lets future agents perfectly infer i 's private signal; there, the public belief updates exactly as the private belief in Equation (13). In the remaining two cases the action carries no information about the true state, so the public belief is unchanged. This yields Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that, for $a \in \{G, B\}$, the probability $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i . It then follows immediately that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

For the second and third cases of Equation (2), it is immediate that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i , for $a \in \{G, B\}$. It remains to treat the first case.

$$\begin{aligned} \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step applies the law of total probability. The second substitutes b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$, using that q_i is a function of $\mathcal{H}_i$. The third uses that $a_i = s_i$ when $1 - q_i \leq b_i \leq q_i$, together with the conditional independence of s_i from the history given ω . The claim follows.

This establishes the following theorem.

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

In other words, b_i is a sufficient statistic for the update of b_{i+1} : it captures all information in $\mathcal{H}_i$ relevant to that update.

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We prove the claim by induction on i .

Base case. For $i = 1$ we have $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. Applying Bayes' theorem gives

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step uses that the unconditioned belief of ω equals the prior b_1 .

Inductive step. Assume the claim holds for all $i \leq n$. Then

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n . This completes the induction and the proof. $\square$

We have thus shown the public belief to be Markovian by adapting the arguments of Banerjee (1992) and Bikhchandani et al. (1992). The adaptation is necessary precisely because the introduction of a central planner alters how agents' precisions are selected, which the classical arguments do not accommodate.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We establish the claim separately for each case of a_i in Equation (2). For the second case,

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, for the third case of Equation (2),

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

Finally, for the case $1 - q_i \leq b_i \leq q_i$, where $a_i = s_i$ by Equation (2),

$$\begin{aligned} \mathbb{P}(a_i \neq \omega|b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a|b_i, q_i) \mathbb{P}(\omega \neq a|b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G|b_i, q_i) \mathbb{P}(\omega \neq G|b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B|b_i, q_i) \mathbb{P}(\omega \neq B|b_i, q_i, s_i = B) \end{aligned} \tag{18}$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B|b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G|b_i, q_i, s_i = B) \end{aligned} \tag{19}$$

$$\begin{aligned} &= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1-q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \end{aligned} \tag{20}$$

$$\begin{aligned} &= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases} \end{aligned}$$

$$\begin{aligned} &= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases} \end{aligned}$$

$$\begin{aligned} &= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases} \end{aligned}$$

Combining the three cases yields the stated result. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Substituting Equation (4) into $r_A(b, q)$ from Equation (5) yields the derivative

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As established in Section 3.2, since social welfare is increasing in precision (Appendix C.9) we may restrict the altruistic planner to precisions $q \in [p, 1]$, with $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) makes p optimal whenever $p < \max(b, 1 - b)$: if the signal is too imprecise to alter the agent's action, the planner declines to expend resources on it. In the complementary case, $\beta(\cdot)$ being increasing and concave makes $\dot{\beta}(\cdot)$ positive and decreasing, so if the optimum lies in this range it must be attained at the endpoint $q = 1$.

It remains to identify the regime in which this latter case is optimal:

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, this inequality can never hold, so precision 1 is never optimal. Otherwise, it holds exactly on $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}\right]$. Together, these two cases establish Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We prove by induction that the expected k -th-stage reward—the planner's expected utility from the control and action of the k -th agent—is convex in public belief. The base case is the instantaneous reward Equation (5), which is itself convex in public belief.

The first difficulty is the unusual structure of the public-belief update: although the state space is uncountably infinite, from any state the update places support on at most two values. To exploit this, we introduce the *decision tree*, the complete binary tree of all trajectories induced by a fixed initial belief and policy. Each node has two children, one per possible signal realization for the next agent; the root is the expected instantaneous reward at time 1, from the first agent's action. The induction descends the levels of this tree, the k -th level comprising 2^{k-1} nodes, each indexed by a length- $(k - 1)$ sequence of signal realizations.

We then show that if a node at level $k - 1$ has convex expected reward, so do its two children at level k . This is exactly where the dependence of agents' actions on the public belief must be confronted: under the same precision and the same signal realization, two agents starting from different public beliefs may take *opposing* actions (see Equation (2)), so the standard tools for bounding future terms of a Markov process do not apply. Here our specific belief update works in our favor. Because Bayesian updates form a martingale ($\mathbb{E}[b_{i+1}] = b_i$), we can combine this martingale property with the convexity of the instantaneous reward to close the inductive step and thereby complete the proof.

Preliminaries We first establish a lemma collecting several useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_0^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned} V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i = 0\}] \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right] \end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now formalize the notion of *decision trees* for Markovian and more general policies. Any policy can be represented as a decision tree, a complete binary tree of the form shown in figure 3. Each node encodes the state of the system—the belief value held when the controller sends its private signal to the agent at that node's level—and the two branches leaving a node encode the possible actions that agent takes upon receiving its signal. The controller's policy (its choice of precision) fixes both the action probabilities at each node and the

resulting posterior beliefs; the action probability at a node is the weight of the corresponding branch. A decision tree is thus uniquely determined by its nodes' belief values and its branches' weights. For a deterministic stationary Markovian policy, the controller's precision—and hence each action probability—is a deterministic function of the node's belief value alone, independent of the node's level and of the path from the root; consequently the entire tree is determined by the initial state. For an arbitrary (non-Markovian) policy, by contrast, the controller's decision at a node may depend on the full path leading to it.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We first show that, at fixed precision, the instantaneous reward r_A is convex in public belief. Using Lemma 7, we rewrite r_A as

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \quad (23)$$

The pointwise minimum of linear functions is concave, and since r_A subtracts this minimum, r_A is convex in x .

Fix arbitrary x_0, m_0, n_0 with $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. Convexity of $V_A^*(\cdot)$ requires $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. Since $V_A^*(x) \geq V_A^\pi(x)$ for every policy π by definition of the optimal value function, it suffices to exhibit a policy $\tilde{\pi}$ for which

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \quad (24)$$

Let Λ denote the decision tree of π^* initialized at state x_0 .

We take $\tilde{\pi}$ to be the controller policy that, at each node, applies the same precision that Λ applies at the *counterpart* node, where two nodes are counterparts when they are reached by the identical sequence of actions from their respective roots (see figure 3). Thus $\tilde{\pi}$ replicates Λ 's precision after any given sequence of agent actions. Denote by Λ_m and Λ_n the decision trees of $\tilde{\pi}$ initialized at m_0 and n_0 , respectively.

We now describe $\tilde{\pi}$ on the trees Λ_m and Λ_n , tracking counterpart nodes and branch weights throughout. Fix a single action path λ by specifying the set $\mathcal{G}$ of epochs at which action G is taken, with $\mathcal{B}$ its complement (see figure 3). In Λ , let x_i be the state at epoch i obtained by starting from x_0 and following π^* along λ ; thus x_i takes one of 2^i possible belief values at epoch i . Analogously, in Λ_m (respectively Λ_n), let m_i (n_i) be the public belief obtained by starting from m_0 (n_0) and following λ , so that m_i (respectively n_i) is the counterpart of x_i . The policy is then specified notationally by $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, and $\tilde{\pi}$ is defined at every node of Λ_m by mirroring the action π^* takes at the counterpart node in Λ . Here $y(x_i, \pi_A^*(x_i))$ (respectively $z(x_i, \pi_A^*(x_i))$) is the weight of the G -branch (B -branch) leaving x_i in Λ ; because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, the weight of the G -branch leaving m_i in Λ_m is $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$, and the analogous identity holds for the B -branch and for Λ_n . The three trees therefore differ in both their node values and their branch weights. Since π^* is deterministic, so is $\tilde{\pi}$; hence, by Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic once the action path $\mathcal{G}$ is fixed.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ denote the expected instantaneous reward at epoch k when starting from state b_0 under policy π , and write $\phi^*(x) = \phi^{\pi^*}(x)$. By Fubini's theorem, the value function decomposes as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

This lets us restate (24) as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1 - t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

which we prove by establishing, for every $i = 0, 1, 2, \dots$, the per-epoch inequality

$$\phi_i^*(x_0) \leq t\phi_i^{\tilde{\pi}}(m_0) + (1 - t)\phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

summing these over i yields (26), hence (24), and finally the convexity of $V^*(\cdot)$ in public belief.

Consider a general epoch $i = k$. The quantity $\phi_{k-1}^\pi(x_0)$ comprises 2^{k-1} terms, one for each action path in $\{G, B\}^{k-2}$. In particular, at epoch $k - 1$ the path λ contributes to $\phi_{k-1}^*(x_0)$ the term

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

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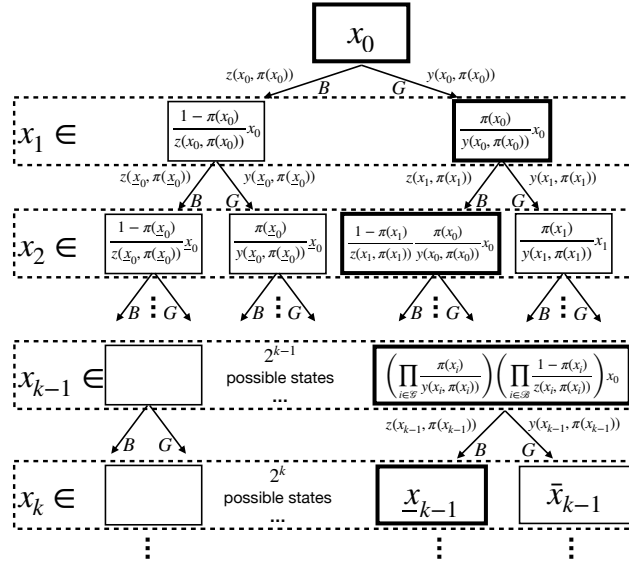
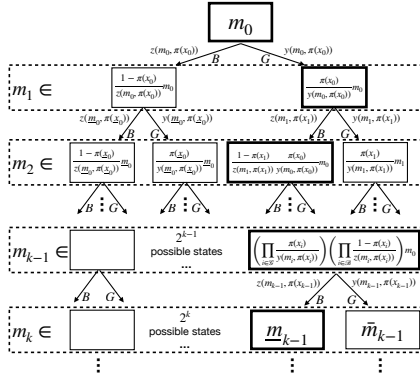
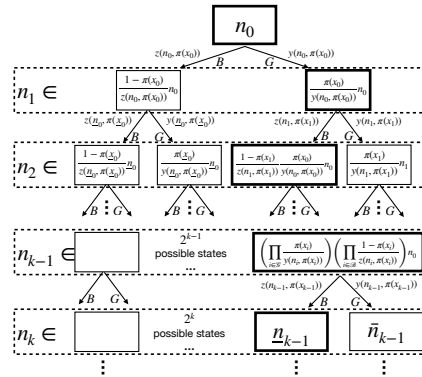
(a) Decision tree Λ rooted at x_0 (b) Decision tree Λ_m rooted at m_0 (c) Decision tree Λ_n rooted at n_0

Figure 3: Belief evolution depicted as three binary trees, rooted at x_0 , m_0 , and n_0 , respectively; level i contains the 2^i states reachable after i epochs. A left branch corresponds to action B , yielding updated belief $\underline{x}_0$ (respectively $\underline{m}_0$ and $\underline{n}_0$), and a right branch to action G , yielding $\bar{x}_0$ (respectively $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with its probability, and the bold states trace an example path λ in which $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states at each level of the three trees are counterparts of one another.

Define $P_\lambda^*(x_0) = (\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i))) (\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)))$, the probability of traversing path λ from initial state x_0 under policy π_A^* . As before, let $\bar{x}_i$ and $\underline{x}_i$ denote the beliefs updated from x_i after observing action G or B , respectively; equivalently, $\bar{x}_i$ and $\underline{x}_i$ are the belief values of the two children (at level $i+1$) of the node x_i at level i . Referring to levels $k-1$ and k in figure 3a, the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ of x_{k-1} contribute to $\phi_k^*(x_0)$ the two terms

$$P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (29)$$

Analogously, define $P_\lambda^{\tilde{\pi}}(m_0) = (\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i))) (\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)))$, the probability of traversing λ from initial state m_0 under $\tilde{\pi}$. Since $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, this equals $P_\lambda^{\tilde{\pi}}(m_0) = (\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i))) (\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)))$, and $P_\lambda^{\tilde{\pi}}(n_0)$ is defined by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms contributed to $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^{\tilde{\pi}}(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & \leq tP_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & + (1-t)P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \end{aligned} \quad (32)$$

The same construction combines the two children of every other term at level $k-1$, producing an analogous inequality for each; summing these inequalities gives (27), and hence (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We focus on the second of the two terms in (29) and establish the two identities

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned} 1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \end{aligned} \quad (34)$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \quad (35)$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \quad (36)$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \quad (37)$$

Applying these to the second term of (29) gives

$$\begin{aligned} & P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\ & \leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\ & \quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{ Eqn 35}) \\ & = t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & \quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37}) \end{aligned} \quad (38)$$

The same argument applied to the first term of (29) yields

$$\begin{aligned} & P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\ & \leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\ & \quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \end{aligned} \quad (39)$$

Summing (38) and (39) yields (32).

It remains to complete the proof of (32)—and thereby the whole proof—by establishing (33) and (34).

Proof of (33)

Iterating the recurrence in Equation (13) and Equation (3), we express $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as products of the successive belief updates applied to x_0 , m_0 , and n_0 , respectively, where each factor is the multiplicative state update induced by a single action:

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \quad (40)$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \quad (41)$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \quad (42)$$

These identities are best read against the illustration in figure 3. Reorganizing the terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, and using $x_0 = tm_0 + (1-t)n_0$, we obtain

$$\begin{aligned} \underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \end{aligned} \quad (43)$$

This proves (33).

Proof of (34)

Consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We prove that it can be written as $\alpha_0 m_0 + \beta_0$ for coefficients α_0, β_0 that depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$, with the precise dependence fixed by the action sequence along this path in stages 1 through $k-1$. The argument is a backward induction on the terms of the product, starting from its final $((k-1)$ -th) term. The base case is this final term, which is either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$ according to the $(k-1)$ -th action:

$$\begin{aligned} y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1})) \end{aligned}$$

Thus the final term has the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$, where α_{k-1} and β_{k-1} depend only on $\pi_A^*(x_{k-1})$ and their form is fixed entirely by the $(k-1)$ -th action (which determines whether the term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, suppose $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$, with $\alpha_{j+1}, \beta_{j+1}$ depending only on $\{\pi_A^*(x_i)\}_{i > j}$ and their form fixed by the actions along this path from $i > j$ to $i = k-1$. We show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can likewise be written as $\alpha_j m_j + \beta_j$, with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ and their form fixed by the actions from $i > j-1$ to $i = k-1$.

Define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$. Thus,

$$\begin{aligned} &\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1}) \end{aligned}$$

Thus, from the definitions of $f(\cdot)$, $g(\cdot)$, $y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, the inductive hypothesis guarantees that $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i>j}$ and the actions along this path from $i > j$ to $i = k - 1$; hence the claim holds for α_j, β_j , completing the induction.

Recall now that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$, which involves the same policy and action sequence as the numerator. Since the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the action path, the identical argument writes the denominator as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 , and likewise handles the numerator of the second term in (34). The coefficients in (33) therefore satisfy

$$\begin{aligned} & t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ & + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ & = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\ & = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = tm_0 + (1-t)n_0) \\ & = 1 \end{aligned}$$

Hence (34) holds, which completes the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Suppose, for contradiction, that there exists $b \in [0, 1]$ with $\pi_A^0(b) > \pi_A^*(b)$. We construct a policy π for which $V_A^*(b) < V_A^\pi(b)$, contradicting the optimality of $V_A^*(\cdot)$.

Let π be the policy that, starting at public belief b , applies precision $\pi_A^0(b)$ at the current step and the optimal policy thereafter. Decomposing $V_A^*(b) - V_A^\pi(b)$ into instantaneous reward r_A and discounted future reward, and writing b^+, b^- for the beliefs updated after signals G, B under the optimal policy and b_m^+, b_m^- for the corresponding updates under the myopic policy, we have

$$\begin{aligned} V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\ &\quad + \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\ &\quad - \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\ &< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\ &\quad - \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\ &\leq 0 \quad (\text{Theorem 2}) \end{aligned}$$

The first inequality uses that, by definition of the myopic optimal policy, the instantaneous reward under $\pi_A^0(\cdot)$ is at least that under any other precision.

The final inequality invokes the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future term: by Equation (13), $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$, and combining this ordering with convexity yields the last step, completing the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ under the policy π that applies precision q in the current step and the optimal policy thereafter; the maximizer q of $V_A^\pi(b)$ is the optimal precision starting from public belief b . Since the planner never benefits from lowering precision below the baseline p (Appendix C.9), we may restrict q to $[p, 1]$ without loss of generality.

Let b^+ and b^- denote the possible upward and downward belief updates after a signal of precision q . For $q \geq \max(b, 1 - b)$, the second derivative of $V_A^\pi(b)$ with respect to q is

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2(1 - b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1 - b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

Given the convexity of $V_A^*(\cdot)$ and the concavity of $\beta(\cdot)$, this expression shows $V_A^\pi(b)$ to be convex in q on $[\max(b, 1 - b), 1]$, so on that interval it is maximized at one of the two endpoints. For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative, so the cost-minimizing baseline p is optimal there. The optimal precision is therefore one of three candidates: p , $\max(b, 1 - b)$, or 1.

The expected value under the latter two candidates is always strictly negative, whereas under precision p it equals 0 at $b = 0$ and $b = 1$. Combined with the non-positivity and symmetry of the value function (Lemma 8), this guarantees a threshold d_A with $\pi_A^*(b) = p$ for $b \in [0, d_A) \cup (1 - d_A, 1]$; moreover, Theorem 1 together with Lemma 9 gives $d_A \in (0, t_M)$, with t_M as in Theorem 1.

Finally, Theorem 1 and Lemma 9 likewise yield a threshold $t_A \in [d_A, t_M]$ with $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$, completing the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Substituting the action rule (2) into the instantaneous reward (6) yields the piecewise form

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Observe that $z(b, q) = b + q - 2bq$ is increasing in q when $b < 0.5$ and decreasing in q when $b > 0.5$.

At $b = 0.5$, we have $z(b, q) = z(0.5, q) = 0.5$ for every precision, and since $q \geq 0.5$ always holds we are in the first case of (44); there the second term of $r_B(b, q)$ is constant in q , so the planner maximizes the first term alone, which occurs at $q = p$.

We now establish a sequence of lemmas that together underpin the proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except at $q = p$, where its derivative with respect to the chosen precision is

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, the second derivative of the reward with respect to q is, for $q \neq p$,

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Since $\beta(\cdot)$ is concave, this is non-negative, so $r_B(b, q)$ is convex in q . The reward is therefore maximized either at one of the endpoints $\{\max(b, 1 - b), 1\}$ or at p , the sole point of non-differentiability. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$ we have $\max(b, 1 - b) = 1 - b > p$.

First suppose $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. By (44), the reward is then maximized by minimizing $|q - p|$, so the optimal precision below $\max(b, 1 - b)$ is p .

Now suppose $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$, and it remains only to rule out $\pi_B^0(b) = 1$. Comparing $r_B(b, 1)$ with $r_B(b, 1 - b)$ —both in the first case of (44)—the increasingness of $\beta(\cdot)$ together with $p < 1 - b \leq 1$ makes the reward's first term larger at precision $1 - b$ than at 1; and since $z(b, q)$ is increasing in q for $b < 0.5$, we obtain $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$. Precision 1 is thus ruled out, completing the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. For any $q \neq p$ the term $-\beta(|q - p|)$ makes the reward strictly negative. For $b \in (p, 1]$ and $q = p$, we have $p < \max(b, 1 - b) = b$, placing us in the third case of (44) with reward 0. Precision p is therefore optimal. $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ reduces to $\{0.5\}$, where $p = 1 - b$ and hence $\pi_B^0(b) = p = 1 - b = 0.5$. We treat $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$, so by the first case of (44) precision p yields reward $r_B(b, p) = -Cz(b, p)$. Consider two subcases, $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

If $q < \max(b, 1 - b) = 1 - b$, the second case of (44) applies with reward $-\beta(|q - p|) - C$. Since $z(b, q)$ is increasing in q for $b < 0.5$ and lies between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$, we have $z(b, q) < 1$ for all q ; hence the reward at any $q < 1 - b$ is strictly below $r_B(b, p)$, ruling out this subcase.

If $q \geq \max(b, 1 - b)$, Lemma 10 leaves candidates $\{1 - b, p, 1\}$. Comparing precision 1 with p , the first term of the reward in (44) is larger at p , and since $z(b, q)$ is increasing in q for $b < 0.5$ the second term is larger at p as well; precision 1 is thus ruled out, completing the proof. $\square$

We now formalize the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

By definition, any optimal policy is ϵ -optimal for every $\epsilon > 0$; in particular $\pi_B^0(b)$ is ϵ -optimal.

Following this definition, we let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or
2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. Consider the two subcases $q \geq \max(b, 1 - b)$ and $q < \max(b, 1 - b)$.

In the first subcase, Lemma 10 gives $\pi_B^0(b) \in \{b, p, 1\}$. Comparing precision b with p , the first term of the reward in (44) is larger at p , and since $z(b, q)$ is decreasing in q for $b > 0.5$ the second term is larger at p too; precision b is thus ruled out, yielding the first claim.

In the second subcase, $q < \max(b, 1 - b) = b$, so (i) $q \in [0.5, b)$ since $q \geq 0.5$, and (ii) the third case of (44) applies, with reward $-\beta(|q - p|)$. Because $0.5 < b \leq p$, this reward is increasing in $q \in [0.5, b)$, which yields the second claim directly. $\square$

Remark 3. As noted in Puterman (1990), Section 2.3.1, an optimal policy may fail to exist over infinite action spaces when the supremum is not attained. In such cases we instead seek ϵ -optimal policies, which yield reward within $\epsilon > 0$ of the supremum; this is precisely the situation in case 2 of Lemma 14.

Proof of Theorem 4 Using $p \geq 0.5$, we partition the analysis into four belief regimes: $b \in [0, 1 - p)$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$. By Lemmas 11–14, the candidate policies in these regimes are $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We establish a threshold t_1 with $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Applying (44), the rewards under the two candidate policies are

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1-b) &= -\beta(1-b-p) - C(1-2b(1-b)) \end{aligned}$$

The derivative of $r_B(b, 1-b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1-b) = \dot{\beta}(1-p-b) + 2C(1-2b)$$

which is positive, since $\beta(\cdot)$ is increasing and $b < 1-p \leq 0.5$. Evaluating at the endpoints $b = 0$ and $b = 1-p$ gives

$$\begin{aligned} r_B(0, 1) &= -\beta(1-p) - C < r_B(b, p) \\ r_B(1-p, p) &= -C(1-2p(1-p)) > r_B(b, p) \end{aligned}$$

Thus $r_B(b, 1-b)$ is increasing on $[0, 1-p]$, with $r_B(b, 1-b) < r_B(b, p)$ at $b = 0$ and the reverse at $b = 1-p$. Since $\beta(\cdot)$ —and hence both reward functions—is continuous, there exists $t_1 \in (0, 1-p)$ with $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1-b$ for $b \in (t_1, 1-p)$.

Case 2: $b \in [1-p, 0.5]$, $\pi_B^0(b) \in \{1-b, p\}$ (Lemma 13)

If $p = 0.5$, the interval reduces to $\{0.5\}$, where $1-b = p$; the two candidates coincide and $\pi_B^0(b) = p$. For the remainder we take $p > 0.5$.

We establish thresholds t_2, t_3 with $\pi_B^0(b) = 1-b$ for $b \in [t_2, t_3]$ and $\pi_B^0(b) = p$ for $b \in [1-p, t_2] \cup [t_3, 0.5]$.

Applying (44), the rewards under the candidate policies are

$$\begin{aligned} r_B(b, 1-b) &= -\beta(p-1+b) - Cz(b, 1-b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1-b$:

$$\begin{aligned} r_B(b, 1-b) &\geq r_B(b, p) \\ \beta(p-1+b) &\leq C[z(b, p) - z(b, 1-b)] \\ \beta(p-1+b) &\leq C(p-1+b)(1-2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1-p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1-p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1-p, 0.75-0.5p]$ and $[0.75-0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1-p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1-p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1-p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1-b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1-b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1-p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b-\epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b-\epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1-p) - C(1-b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b-\epsilon) &= -\beta(p-b+\epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b-\epsilon)$ is arbitrarily close to 0, and, therefore, $b-\epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b-\epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b-\epsilon) - r_B(b, p)$ and $r_B(b, b-\epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b-\epsilon$ is the optimal policy, must also be convex. Therefore, $b-\epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. *For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,*

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. *Biased value function continuity*

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1-b, 1, b-\epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1-b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of (44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum,

$r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ +\delta y(b, q)V_B^*(f(b, q, G)) \\ +\delta z(b, q)V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ +\delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1 - \delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1 - \delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum

control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1 - b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q - p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1 - b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1 - \epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q - p|) \rightarrow \beta(1 - p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p - p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1 - p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1 - p, 0.5)$, $\pi^*(b_i) \geq 1 - b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1 - b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1 - b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1 - \delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1 - \delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1 - b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1 - b_i$.

The remaining candidate policies are $q_i \in [1 - b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1 - b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1 - b_i)$ and $b \in [1 - p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1 - q_i)}{z(b_i, q_i)}$. Both of these expression are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1 - b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1 - b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$,

$\mathbb{P}(a_i = G|b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^*(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b]} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \quad (57)$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \quad (58)$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. □

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

Here we argue that, for the altruistic planner, the problem we consider is equivalent to one that involves two types of individuals with differing preferences.

Assume that there are two types of agents G and B with the type of agent i denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise. The second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Further, assume that all agents know their own type and the types of all other agents. All other aspects of the model remain the same.

The private belief update of agent i can be expressed as below. This remains unchanged because it only depends on the signal structure, i.e., $\mathbb{P}(s_i = \omega) = q_i$.

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_iq_i-b_i-q_i} b_i & s_i = G \\ \frac{1-q}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows with superscript indicating the type of agent i and $!$ indicating negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because we assume the type of each agent is known to all, when $q_i \geq \max(b_i, 1 - b_i)$, all agents can perfectly infer the private signal of agent i . If $q_i < \max(b_i, 1 - b_i)$, the action of agent i is uninformative.

Thus, after observing agent i 's type t_i , precision q_i , and action a_i , the public belief is updated as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting, an altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$ as these choices align with the personal utility functions of the agents.

Thus, we can express the planner's instantaneous reward as follows with superscript denoting the type of agent i :

$$\begin{aligned} r^G(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \\ r^B(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \end{aligned}$$

These are identical and equal to the reward function in Equation (5), leading to the equivalence of the altruistic planner's problem when we have heterogeneous but observable types. ■

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate the controlled social learning setting, we consider a sequence of agents deciding whether or not to buy a new model of car informed by observations of past actions and an ad of precision selected by the planner. LLMs served three distinct roles: (1) the agents, (2) the planner, and (3) an oracle. Each agent was assigned a 'profile' containing demographic information and their car use cases and preferences. At the onset, we select the good or bad car as the offering. The information sets and objectives of the planner and agents were exactly as in the model of Section 3.

The oracle's role was two-fold: (1) it generated a generic message to serve as the baseline precision signal without intervention by the planner, and (2) once the planner chose a precision, it generated a message with the prescribed precision, customized to the profile of the target agent.

The oracle was provided with a fact sheet (safety rating, customer satisfaction, etc.) for each type of car along with the agents’ profiles. Together, the fact sheet and profile facilitated the generation of a tailored message with precision prescribed by the planner. More detail on the experimental setup and prompting can be found in Appendix E.2.

Each agent was tasked with making a binary decision based on observations of previous decisions and their private signal. The agents’ goal was to match the true parameter value.

The planner was provided the same decision history and tasked with choosing a precision level for each agents’ message. They were also given the precision of a generic, un-tailored message which they could send for free. In the altruistic case, the planner’s objective was to induce correct decisions, while in the biased case, their goal was to induce agents to buy the car.

We compare the LLM simulations to the analytic solutions obtained via JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

Experiments were run using the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run the LLM simulations for the parameters in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Instance parameters

The car facts were designed so that the good car was good for everyone and the bad car was bad for everyone. Below we include the fact sheets for the GOOD and BAD cars, respectively.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
    "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
    "No factory towing approval.",
    "Average 3.8 service visits per year (battery-alert issues).",
    'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
    "Not eligible for federal tax credit.",
    "OTA updates limited to infotainment."
]
```

Profiles were generated programmatically. Profiles included an agent’s name, age, city of residence, interests, planned use case for the car (e.g., commuting, road trips, etc.), and major pain points or concerns (e.g., fuel efficiency, winter driving, etc.).

```

1836 E.2.2 PROMPTS
1837
1838 Agent LLM Prompt:
1839
1840 sys = (
1841     f"You are deciding whether to buy a car. "
1842     f"Your profile is {profile}"
1843     f"You start believing there's a 50% chance the car is GOOD. "
1844     f"You have two information sources: "
1845     f"1. Previous agents' decisions "
1846     f"2. An advisor's message about the car "
1847     f"Your goal is to maximize your reward "
1848     f"i.e. BUY if the car is GOOD and "
1849     f"NOT BUY if the car is BAD. "
1850     f"If you BUY a GOOD car, you GAIN ${C}. "
1851     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1852     f"If you BUY a BAD car, you LOSE ${C}. "
1853     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1854     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1855     f"decision = exactly 'BUY' or 'NOT BUY'."
1856 )
1857 user = (
1858     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1859     f"ADVISOR MESSAGE:\n{message}\n\n"
1860     f"TASK: Use the previous decisions and the advisor message. "
1861     f"Decide BUY or NOT BUY. "
1862     f"Update your belief about car quality (0.0-1.0). "\n"
1863     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}}'
1864 )
1865
1866 Planner LLM Prompt:
1867
1868 sys = (
1869     f"You are providing messages to potential car buyers. "
1870     f"Each potential car buyer (or agent) has the same information. "
1871     f"Your goal is to get agents to {objective}."
1872     f"Each decision according to your objective, earns you {C} dollars. "
1873     f"You and the agents have the same information. "
1874     f"Initially there is a 50% chance the car is GOOD. "
1875     f"The agents' aim to BUY if the car is GOOD "
1876     f"and NOT BUY if the car is BAD. "
1877     f"You can send a baseline message for free "
1878     f"or tailor the message to the agent at a cost. "
1879     f"Tailoring costs {k} dollars times the change in precision. "
1880     f"(Message precision is the probability (0.5 to 1.0) "
1881     f"that the message leaves the buyer with the correct impression. "
1882     f"For example, if the car is likely GOOD: "
1883     f"a high precision message will likely suggest BUY "
1884     f"and a low precision message will likely suggest NOT BUY. "
1885     f"You may choose to increase or decrease precision. "
1886     f"Agents decide based on your message and the decision history. "
1887     f"They update beliefs by observing previous decisions. "
1888     f"IMPORTANT: Thus your objective is to maximize:"
1889     f"{C} * (# of agents who choose according to your objective) "
1890     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1891     f"where {d} is your discount factor for future cost. "
1892 )
1893 user = (
1894     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1895     f"DECISION HISTORY:\n{history_txt}\n"
1896     f"TASK:"
1897     f"Based on the history and your objective, "
1898     f"choose your desired precision.\n"
1899     f'JSON response: {{ "message_precision": <number> }}\n'
1900 )

```

Oracle LLM Prompt:

```

sys = (
    f"You will be given "
    f"1. an agent profile, "
    f"2. information about a car, "
    f"3. the true quality of the car, "
    f"4. a baseline message and the 'precision' of that message, and "
    f"5. a desired precision for the message you output. "
    f"(Message precision is the probability (0.5 to 1.0) "
    f"that the message leaves the buyer with the correct impression. "
    f"For example, if the car is likely GOOD: "
    f"a high precision message will likely suggest BUY "
    f"and a low precision message will likely suggest NOT BUY. "
    f"IMPORTANT: Output a message with the requested precision. "
    f"IMPORTANT: The message should be tailored to the profile. "
)
user = (
    f"AGENT PROFILE:{profile}\n"
    f"TRUE CAR QUALITY: {omega}\n\n"
    f"CAR FACTS: {car_facts}\n"
    f"BASELINE MESSAGE:{baseline_message}}\n"
    f"BASELINE MESSAGE precision: {p}\n"
    f"DESIRED MESSAGE precision: {q}\n"
    f"IMPORTANT: Output a message with the requested precision. "
    f"TASK: Write a message (MAX 100 words) with the desired precision."
    f'JSON response: {"message": "<message>"})'
)

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We now provide further experiments to validate the self-reported beliefs of the LLMs (as shown in figure 1b) and the performance of the LLM oracle. In these experiments, we utilize the LLMs as rational and utility maximizing, but not necessarily Bayesian, agents. The goal of these experiments is (1) to demonstrate that the model's self-reported posterior genuinely corresponds to the belief that governs its choices in state-contingent decision problems, and (2) to verify that messages produced by the oracle do have the desired precision.

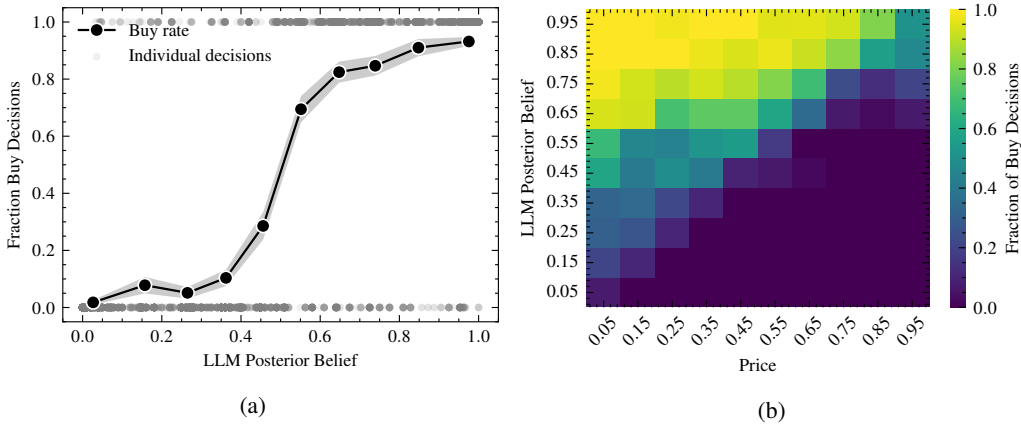


Figure 4: The rate at which the LLM chose a ‘buy’ decision as a function of (a) its self-reported posterior belief (b) its self-reported posterior and the price buying.

First, we provide the LLM with a prior and a new signal, obtain its self-reported posterior belief, and then ask it to make a binary “buy/not-buy” decision that yields unit reward if and only if the true state is good. A rational agent should choose “buy” exactly when its subjective belief exceeds 0.5. As shown in figure 4a, the LLM’s decisions align sharply with this threshold: it overwhelmingly chooses “buy” when its reported posterior exceeds 0.5 and “not buy” otherwise. Next, we elicit the model’s willingness to pay for a payoff that is contingent on the true state. For a range of prices c , we ask whether the agent would pay c in exchange for receiving unit utility if the state is good. A rational agent with belief p should accept exactly when $c <$

p . Figure 4b plots the fraction of “buy” decisions across posterior–price pairs, showing a clear separation: acceptance rates are high when $c < p$ and low when $c > p$, precisely as expected from a utility-maximizing decision maker whose belief is p . Both experiments demonstrate that the self reported posterior is close to the inherent belief of the model as used in decision making.

Similarly, we validate the performance of the oracle in crafting messages of a given precision. After the oracle generates a message of specified precision q , we ask another LLM to form an estimate $\hat{q}$ of the message’s precision. We do this for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ on 100 programmatically generated agent profiles with 10 trials for each combination for a total of 6,000 trials. The mean absolute error $|q - \hat{q}|$ was just 0.025. This indicates that the realized precision of the oracles message’s does closely follow the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We now unpack belief trajectories from an example instance of the LLM experiment to elucidate the impact of non-Bayesian belief updating. Here we fix the policy to be the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly towards 0. The LLM belief (solid black line), however, is more “stubborn.” This macro-level trajectory is a direct result of the micro-level biases: the public belief resists entering a cascade (NB3) because individual agents overreact to the occasional positive signal that runs counter to the group’s increasingly negative belief (NB2).

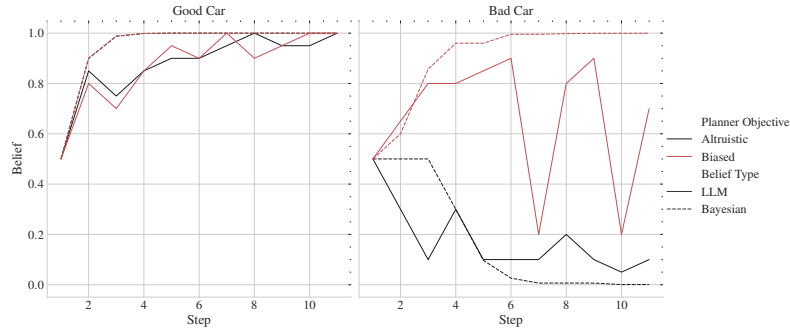


Figure 5: Agent Belief Updating trajectories for the true Bayesian belief and the belief reported by the LLM agent under analytically optimal policies for different planner objectives and true states of the world.